“SMARTCINNAMON”: A MULTI-MODAL MACHINE LEARNING AND IOT FRAMEWORK FOR PRECISION CINNAMON CULTIVATION

RP-25-26J-375

Project Proposal Report

Randeni R.P.R.M

B.Sc. (Hons) Degree in Information Technology

Specializing In Information Technology

Department of Information Technology

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DECLARATION

We declare that this is our work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Sarigeevi Date 28/08/2025

Signature of the Co - supervisor: Date

ABSTRACT

Cinnamon Leaf Fungal Spread Predictor

Ceylon cinnamon is a high-value export crop in Sri Lanka, yet fungal leaf infections significantly reduce yield quality and plantation productivity. Existing solutions primarily detect infections at their current stage but fail to provide early warnings about disease progression, limiting farmers' ability to take timely preventive measures. This study proposes the Cinnamon Leaf Fungal Spread Predictor, a mobile-based system integrated into the SMARTCINNAMON framework. Using smartphone-captured images, the system preprocesses leaf data through resizing, denoising, and segmentation to isolate infected regions. Annotated images at multiple infection stages form a temporal dataset to train Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models for time-series forecasting. Given a single mobile image, the model predicts both the current infection percentage and the expected infection spread over a defined period (e.g., three days). Predictions are visualized through a mobile app interface, which also provides actionable recommendations such as spraying, pruning, or isolating infected plants. Anticipated results include improved early detection, reduced yield losses, and enhanced plantation health through proactive disease management. In conclusion, the Cinnamon Leaf Fungal Spread Predictor introduces a novel, data-driven, and farmer-friendly solution with significant potential to transform sustainable cinnamon cultivation practices.

Keywords: Cinnamon, Fungal Infection, Disease Prediction, Image Processing, Time-Series Forecasting, LSTM, GRU, Mobile Application

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LIST OF ABREVATIONS

Table I: List of Abbreviations

GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
ML	Machine Learning
RMSE	Root Mean Square Error
SDLC	Software Development Life Cycle
API	Application Programming Interface
IDE	Integrated Development Environment
LKR	Sri Lankan Rupee
MAE	Mean Absolute Error
WBS	Work Breakdown Structure
ML	Machine Learning

1. Introduction

Ceylon cinnamon is one of Sri Lanka's most valuable export crops and a significant contributor to the country's economy. However, fungal leaf infections pose a serious threat to cinnamon plantations, leading to reduced yield quality, lower productivity, and financial losses for farmers. Traditionally, farmers rely on visual inspections and manual monitoring to identify infected leaves, but these methods are often time-consuming, inaccurate, and reactive. By the time visible symptoms are detected, the infection has typically progressed to critical stages, making control more difficult and increasing the chances of severe crop damage.

Current disease detection techniques primarily focus on identifying infections that already exist rather than predicting their future spread. These conventional systems lack predictive intelligence and fail to provide early warnings, leaving farmers with limited time to take preventive measures. Consequently, fungal infections continue to spread rapidly, especially under favorable environmental conditions such as high humidity and rainfall. This limitation has led to significant economic losses in the cinnamon industry, creating an urgent need for smarter, data-driven solutions that go beyond basic detection.

Recent advances in machine learning (ML) and mobile-based computer vision have created new opportunities for enhancing plant disease management. By analyzing time-series image data, modern forecasting models such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) are capable of learning infection growth patterns and predicting how diseases will progress over time. For example, Hasan et al. applied a CNN-LSTM architecture to detect spatiotemporal water stress progression in chickpea plants and achieved an accuracy above 97 percent, showing the effectiveness of combining convolutional and recurrent models for plant stress forecasting [1]. Similarly, Yao *et al.* proposed a spatiotemporal predictive learning framework (DPP) to forecast the progression of Ginkgo leaf blight using image sequences, significantly improving prediction performance compared to prior models [2]. These studies demonstrate the potential of deep learning for progression forecasting; however, most existing agricultural solutions are generalized across multiple crops and do not provide cinnamon-specific fungal spread predictions.

This gap highlights the necessity of developing a dedicated mobile-based framework to address the unique environmental and biological challenges of cinnamon cultivation. To overcome these challenges, this study proposes the Cinnamon Leaf Fungal Spread Predictor, a mobile-based system integrated into the SMARTCINNAMON framework. The proposed solution uses smartphone-captured images of cinnamon leaves to preprocess, analyze, and predict fungal spread over a defined future period, such as three days. The system applies LSTM/GRU models to forecast infection progression and visualizes the predictions through a user-friendly mobile app. In addition, the app provides actionable recommendations such as spraying, pruning, and isolating infected plants, helping farmers take preventive measures at an early stage. This proactive approach empowers farmers to manage fungal infections effectively, minimize crop losses, and improve the long-term sustainability of cinnamon cultivation.

1.1 Background & Literature Survey

The increasing integration of Artificial Intelligence (AI) and Machine Learning (ML) into agriculture has transformed traditional farming practices, particularly in the detection and management of plant diseases. Ceylon cinnamon, one of Sri Lanka's most valuable export crops, faces a significant threat from fungal leaf infections, which reduce yield quality, downgrade export value, and negatively affect farmer income. These infections spread rapidly under humid and rainy conditions, and traditional manual inspection methods fail to detect the disease at early stages, resulting in severe losses. The introduction of mobile-based disease detection and predictive analytics offers an opportunity to mitigate these issues by enabling early detection and forecasting of disease progression.

Several studies highlight the effectiveness of AI-powered image processing in agriculture. For example, convolutional neural network (CNN)-based models have achieved high accuracy in detecting plant diseases using leaf images [3]. These models analyze captured images, classify diseases, and assist farmers in early-stage management. However, such systems are limited to current-state detection and do not provide insights into how infections evolve over time, leaving farmers unprepared for future outbreaks.

Recent advancements in time-series deep learning techniques such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models have opened new possibilities in predicting agricultural phenomena. For instance, studies on grapevine mildew prediction demonstrated that LSTM-based models can accurately estimate disease spread patterns over time [4]. Applying similar models to cinnamon fungal infections would allow forecasting of infection growth and support proactive interventions.

Mobile technology has become increasingly important in precision agriculture. Applications enabling farmers to capture leaf images and receive automated disease detection have significantly improved accessibility to diagnostic tools [5]. However, most existing mobile solutions lack predictive capabilities and focus only on identification, making them less effective for preventive decision-making. A cinnamon-specific mobile application that integrates image-based detection with time-series forecasting models bridges this gap, providing actionable insights and improved disease management.

Moreover, IoT-enabled smart farming solutions combined with mobile-based disease prediction have gained traction in recent years. Studies show that environmental parameters such as humidity, temperature, and rainfall have a strong influence on the spread of fungal infections [6]. Integrating these environmental factors with predictive models significantly enhances accuracy, enabling farmers to make more informed and timely decisions.

The proposed Cinnamon Leaf Fungal Spread Predictor builds on these advancements by introducing a mobile-based predictive framework that not only detects fungal infections but also forecasts their potential spread using smartphone images and LSTM/GRU-based time-series models. Predictions are delivered via a mobile application, along with actionable recommendations such as pruning, spraying, or isolating infected leaves. By combining mobile-based computer vision, time-series modeling, and environmental awareness, this system addresses a critical gap in proactive cinnamon disease management.

Figure 1 below illustrates an example of capturing cinnamon leaf images via mobile for disease detection and predictive analysis.

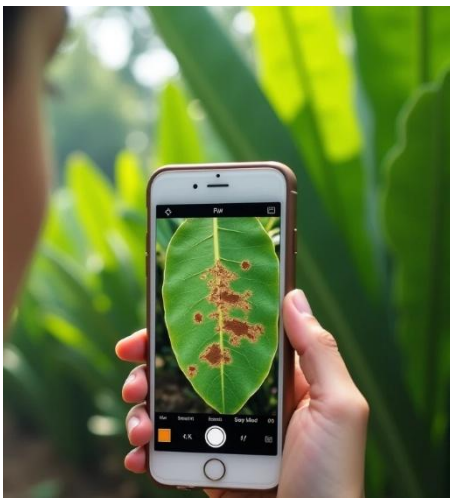

Figure 1:Smartphone-based image capture for cinnamon leaf fungal prediction

1.2 RESEARCH GAP

Fungal diseases in cinnamon leaves are still an issue for Sri Lanka's spice industry, yet technologies have been more aimed at real-time identification rather than predictive future spread. Most recently, deep learning models such as YOLOv5 and CNN-based applications have been used on cinnamon leaves, with recent studies indicating moderate accuracy for leaf spot and canker identification and enabling mobile-based diagnosis. Similarly, widely used smartphone applications like Plantix, Nuru, and Crop Doctor provide image-based classification and basic treatment recommendations. These approaches are, however, limited to the diagnosis at a single time-point and give farmers an instant snapshot of disease severity today without advice on how the disease might progress in the days and weeks that follow.

One of the key flaws in existing plant disease detection pipelines is their reliance on static images to draw conclusions regarding present infection status. While effective, this approach has no regard for the temporal nature of disease development. In other plants, research has shown that incorporating sequential data dramatically improves accuracy. For example, GRU-type time-series models for satellite images have correctly forecast soybean sudden death syndrome development with better accuracy and the ability to demonstrate that forecasting of progression is possible. These findings point to the potential of temporal models such as RNNs, LSTMs, and GRUs for forecasting the dynamics of plant diseases. Yet, these types of applications remain unknown to a great degree for cinnamon, especially at the scale of individual leaves from mobile imagery.

Equally important is the lack of a dataset for cinnamon that records the progression of fungal infections over time. Current studies on cinnamon focus on detection and staging severity but do not attempt to monitor or model the speed at which leaf spots expand. This gap is mirrored in current mobile apps, which provide only instantaneous symptom recognition and not predictive feedback. Therefore, cinnamon farmers receive no advance notice that would allow them to spray, prune, or quarantine infested plants before the disease becomes unmanageable.

Generally, while machine learning advancements have enhanced real-time detection availability, there is no single cinnamon mobile solution that forecasts fungal spread in the long term. Addressing this discrepancy through temporal dataset development and application of sequence models would allow for predictive frameworks to deliver prompt actionable warnings to farmers, consequently reducing yield loss and enabling more sustainable disease control.

Table II: Research Gap

Conducted By	**Cinnamon-Specific Dataset**	**Image-Based Detection**	**Predictive Forecasting (LSTM/GRU)**	**Environmental Factor Integration**	**Mobile-Based Solution**
Research A [7]	✓	✓			✓
Research B [8]		✓			✓
Research C [9]		✓	✓		
Research D [10]				✓	
Research E [11]		✓			✓
Research E [12]		✓	✓		
Proposed System	✓	✓	✓	✓	✓

1.3 RESEARCH PROBLEM

Cinnamon leaf fungal infections progress rapidly under conducive field conditions, yet most available mobile tools and research prototypes only detect the current disease state from a single image. Farmers therefore lack actionable, forward-looking insight specifically, a quantified estimate of how much the infection is likely to spread in the coming days. Without this capability, preventive actions (timely pruning, targeted spraying, isolation of affected leaves/branches) are delayed, causing unnecessary yield loss and quality degradation. The proposed Cinnamon Leaf Fungal Spread Predictor targets this gap by forecasting the future infection percentage from a mobile photo (optionally enriched with past observations), and presenting clear, early recommendations.

1.3.1 Absence of Forecasting for Infection Progression

Current mobile disease apps and many academic models emphasize single-timepoint classification (healthy vs. infected or disease type), not progression forecasting. Farmers can see that a leaf is infected, but cannot estimate whether an infection covering, for example, 18% today will likely reach 35–40% within three days. This missing forecast prevents proactive intervention planning (chemical/biological control timing, canopy sanitation, harvest adjustments). The research problem, therefore, is to design a time-series prediction pipeline. fed by smartphone imagery that outputs infection percentage at user-selected horizons (e.g., +3 days), with uncertainty bounds suitable for field decision-making.

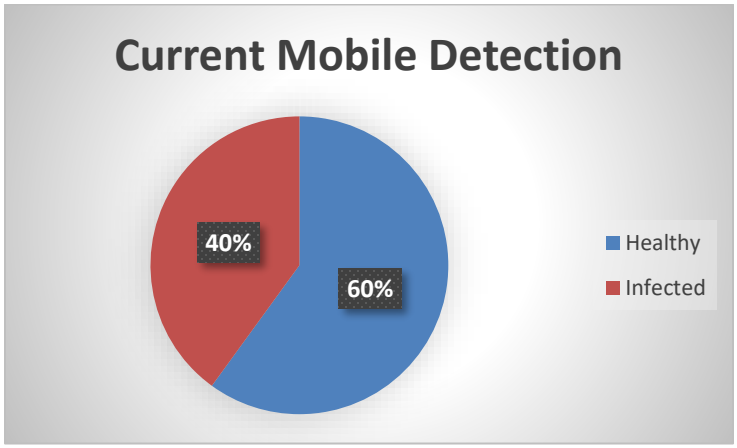

Figure 2:Current Mobile Plant Disease Detection

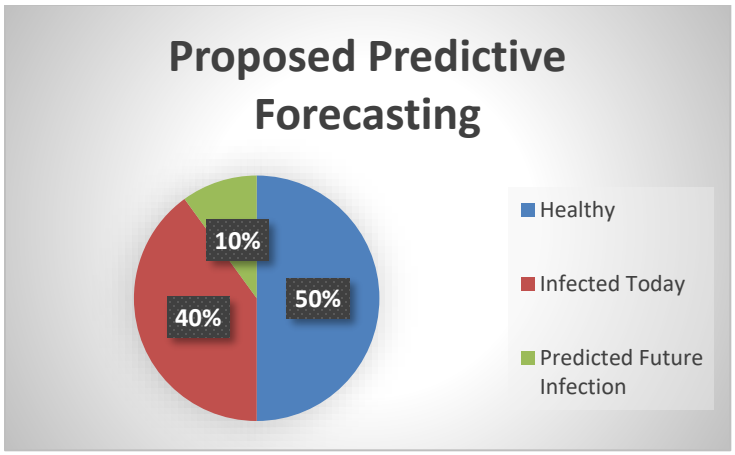

Figure 3: Proposed Predictive Forecasting

1.3.2 Lack of Cinnamon-Specific Longitudinal Data and Ground Truth

Progression models require temporal datasets of the *same* leaf or plant across time, with reliable ground-truth infection area. For cinnamon, such datasets are scarce: images are typically cross-sectional, labels are coarse (diseased/not), and infection masks/percentages are not annotated. This creates two methodological challenges:

- Data creation: capturing serial images of infected cinnamon leaves across stages, under realistic mobile conditions (variable lighting, angles, occlusions).
- Labeling: establishing consistent infection-area percentages (via segmentation + quality-controlled annotation) and accurate timelines.
  Solving this requires a practical protocol for image preprocessing (resize, denoise), segmentation of infected regions, semi-automated mask refinement, and time-indexing of sequences to enable LSTM/GRU training.

1.3.3 Limited Accessibility and Actionable Guidance for Smallholders

Systems that attempt disease forecasting often depend on specialized sensors or dense IoT setups, which are costly and connectivity-dependent barriers for smallholder cinnamon farmers. Even when predictions are produced, outputs are rarely translated into clear, field-ready actions. The problem, therefore, is twofold:

1. deliver lightweight, on-device or low-bandwidth inference from standard mobile photos, and
2. pair forecasts with concise recommendations (e.g., “Likely +15–20% spread in 3 days → prune affected leaves within 24 h; apply recommended fungicide X at dose Y; reassess with a follow-up photo on Day 3”).

2. OBJECTIVE

2.1 Main Objectives

The primary objective of the work in this presentation is to create a mobile-based fungal disease predictor system for cinnamon leaves using a high-level image processing and time-series machine learning technique. Unlike the traditional tools, which detect infections only at the current time, the system has the ability to forecast future instances of fungal disease, empowering farmers with forearmed decision-making capability. With the addition of infection detection through computer vision and spread prediction through LSTM/GRU, the above solution provides early warning of the imminent crop destruction. Lastly, the goal is to create a low-cost, farmer-friendly mobile application that not only detects the stage of the infection but also its future development, allowing farmers to begin prevention processes such as spraying, pruning, or quarantining affected plants to minimize yield loss and attain maximum sustainable cinnamon yield.

2.2 Specific Objectives

2.2.1 Dataset Creation and Preprocessing for Infection Analysis

The goal here is to collect and preprocess the images of the cinnamon leaves at different infection levels. Images collected will be resized, denoised, and segmented to bring out infected regions. Infection levels and timeline annotated data will be considered when developing the foundation for successful model training. This is so that the system has a standard and representative dataset for fungal infection growth.

2.2.2 Feature Extraction and Temporal Data Modeling

The goal here is to extract infection features from the pre-processed images and transform them into time series. With a time-series dataset, the system will be simulating the dynamic temporal growth pattern of the fungi. The features will then be effective inputs for machine learning models with learning capabilities that possess the capacity to recognize infection growth patterns through time.

2.2.3 Creating Predictive Machine Learning Models

This entails building, training, and validating complex time-series forecasting models such as LSTM and GRU. These will forecast the future percentage spread of the fungal disease based on history image data and a mobile input. Deep learning algorithms enable the system to make the correct predictions for a given time frame (e.g., three days).

2.2.4 Farmer Usability Mobile Application Implementation

This is with a view of developing a mobile application that integrates the trained ML models to provide the predictions to the end-users directly. The application will show the current infection rate, forecast the expected rate of spread, and provide real-time treatment recommendations. The interface is supposed to be user-friendly enough to guide farmers with varying degrees of technical know-how.

2.2.5 System Evaluation and Validation

The reason for doing this is to determine the validity and reliability of the fungal spread predictor from the model performance measures such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score. Validation in real life will also be carried out by implementing the system in cinnamon estates to verify the usability, feasibility, and effectiveness of the system in reducing crop loss

3. METHODOLOGY

3.1 SYSTEM OVERVIEW DIAGRAM

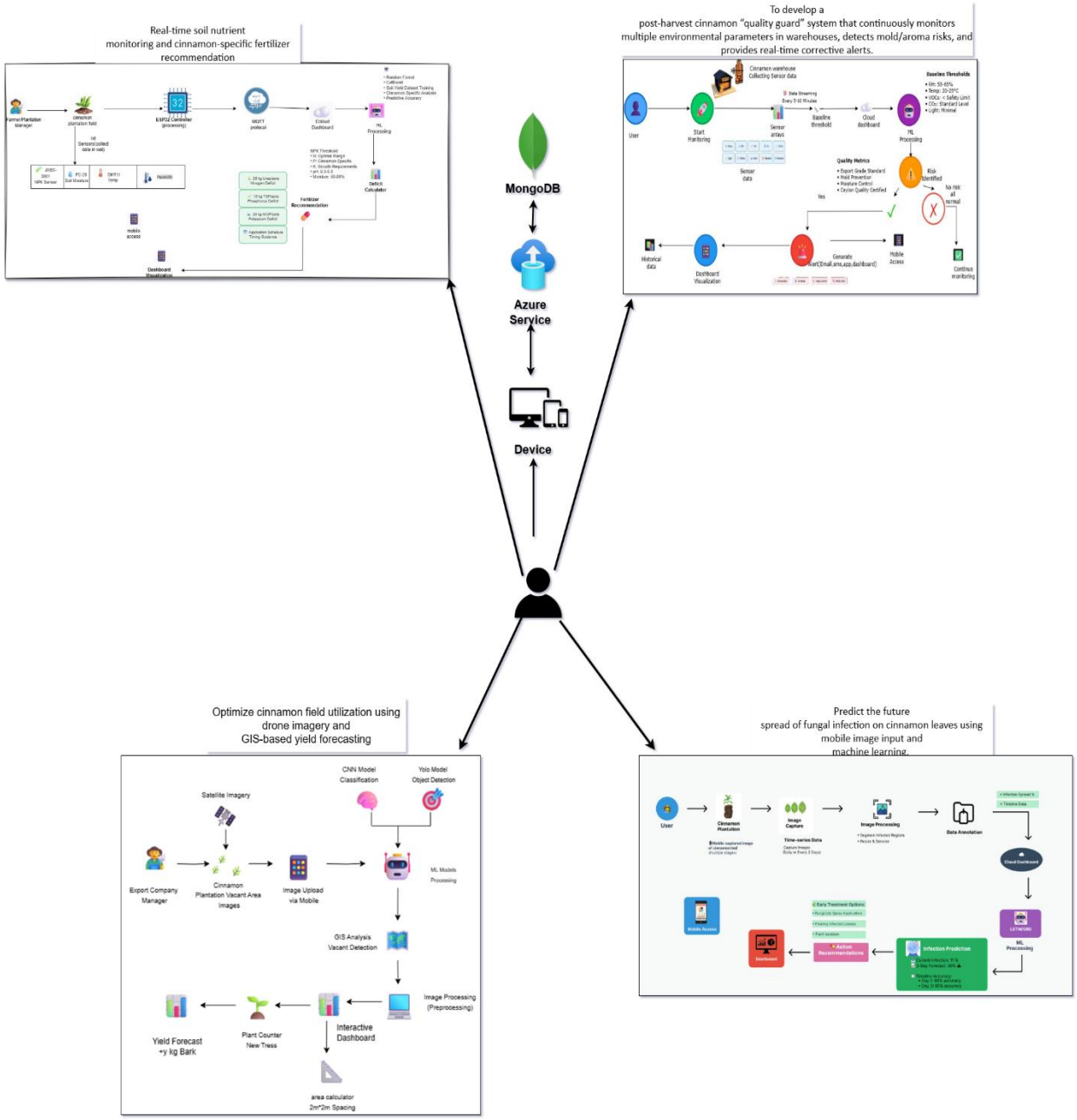

Figure 4:System Overview Diagram

SMARTCINNAMON is an integrated AI–IoT–GIS platform for the cinnamon value chain. Field and warehouse devices stream data to Azure services and MongoDB; machine-learning models turn that data into real-time recommendations, risk alerts, forecasts, and dashboards delivered on mobile and web.

At the center is the user's device (mobile/web) connected to cloud services. IoT sensors in fields and warehouses send readings to Azure (via MQTT/REST) and are stored in MongoDB. Cloud ML services analyze the incoming data continuously and return insights to dashboards and alerting channels. The platform combines four modules that work together to raise yield, protect post-harvest quality, and control disease.

Real-time Soil & Fertilizer Intelligence - ESP32 nodes read NPK (JXBS-3001), soil moisture (FC-28), and temperature/humidity (DHT11). Live values are compared with cinnamon-specific target ranges. ML models (e.g., Random Forest, CatBoost) learn from soil + yield history to recommend exact fertilizer doses (e.g., kg/acre), showing a percent "deficit" for N, P, and K and converting that into actionable inputs.

Post-harvest "Quality Guard" - Multi-sensor stations monitor 10 factors inside warehouses: Temperature, Relative Humidity, CO_2, O_2, VOCs, light intensity, airflow, dust/particles, vibration/shock, and surface moisture. Baselines (e.g., RH 55–65%, 20–25 °C) and ML analytics detect mold/aroma risk early, trigger corrective guidance (ventilate, dehumidify, reduce light), and keep audit-ready logs and trends.

GIS Field Planner & Yield Forecast - Drone/satellite imagery is segmented to map **cultivated vs. vacant** areas. The system proposes optimal replanting layouts using standard spacing (e.g., 2 m × 2 m), estimates additional plant counts, and combines tree counts with historical bark productivity to forecast yield. Results appear on an interactive spatial dashboard: "X new trees → +Y kg expected yield."

Leaf Disease Detection & Spread Prediction - A mobile photo of a cinnamon leaf is analyzed by a CNN to detect current infection and severity. Time-series models (LSTM/GRU), trained on annotated progressions, predict how the infection will grow over the next few days (e.g., "+18% in 72 h"). The app visualizes the forecast and recommends early actions such as pruning, targeted spraying, or isolating plants.

Data Flow - Sensors/Images → Azure ingest → MongoDB storage → ML analytics → Dashboards & Alerts → Farm/warehouse actions.

Value - The platform delivers precise fertilizer dosing, proactive post-harvest quality control, optimized land utilization with yield gains, and early disease intervention—improving productivity, compliance, and sustainability across the cinnamon supply chain**.**

3.2 COMPONENT SYSTEM DIAGRAM

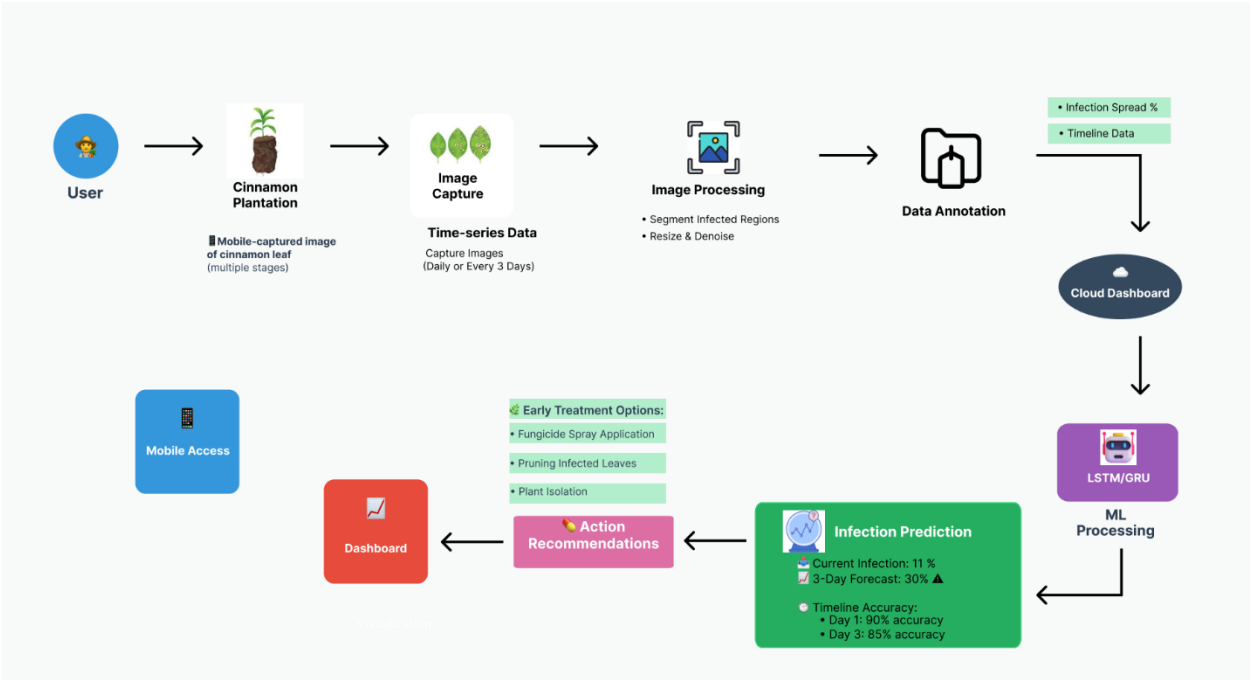

Figure 5:Component System Diagram

According to the system architecture, users can monitor and predict the spread of fungal infection on cinnamon leaves through a mobile application that uses artificial intelligence. The system is designed to be intuitive and accessible for all levels of cinnamon farmers, especially those with limited technical experience. It simplifies the process of identifying, tracking, and managing plant diseases by combining smartphone-based image capture with time-series machine learning techniques.

The first step of the process begins in the field. Farmers use their mobile devices to take clear photographs of cinnamon leaves suspected to be infected. These images are collected at different time intervals.such as daily or every few days.allowing the system to observe how the infection progresses over time. This method ensures that even subtle changes in the infection stage are captured and used for analysis.

After image capture, the system automatically enhances the quality of the photo by reducing background noise, resizing it for consistency, and identifying the infected regions of the leaf. This makes it easier for the system to focus only on the relevant parts of the image. Each photo is then labeled with an estimated percentage of infection and the date it was taken. These images and labels are stored securely in a cloud system, which keeps a growing timeline of each plant's condition.

The core of the system uses this timeline data to predict how the infection will spread in the coming days. This is done through a specially trained AI model that has been designed to learn from past patterns of fungal spread. By comparing current images with historical data, the model can accurately forecast how much of the leaf is likely to be infected after a certain number of days.such as a 3-day forecast. These predictions are automatically calculated and delivered back to the farmer through the app.

Once the prediction is made, the app provides easy-to-understand visuals and recommendations. For example, it may show that the current infection level is 11% but could rise to 30% within three days if no action is taken. To help prevent this, the system suggests specific treatments such as spraying fungicide, pruning the infected leaves, or isolating the affected plant. These suggestions are tailored to the farmer's real-time situation and provided in simple language to guide quick action.

The mobile application is designed to function even in rural environments with limited connectivity. It presents results through a clean interface with icons, percentages, and action buttons that anyone can understand. This ease of use ensures that even farmers without technical knowledge can benefit from advanced technology.

The Cinnamon Leaf Fungal Spread Predictor is not just a disease detection tool.it is a smart assistant for early intervention. By predicting how a fungal infection will grow, it helps farmers act before serious damage occurs. This reduces crop loss, improves long-term yield, and supports more sustainable and profitable farming. The system's combination of accessibility, accuracy, and automation makes it a valuable tool for modern cinnamon cultivation.

3.3 PROJECT TECHNOLOGY STACKS

Table III: Project Technology Stack

Tools & Technologies	Purpose
React-Native	For developing the mobile application frontend.
Python	For image preprocessing, feature extraction, and machine learning model implementation.
TensorFlow / Keras/PyTorch	For training and deploying deep learning models (LSTM/GRU).
MongoDB	Use As Database
OpenCV	For image preprocessing tasks such as resizing, denoising, and infected area segmentation.
Visual Studio Code	Use As IDE
Git	Use Git as the version controller.
Postman	Use postman for test API requests.
Matplotlib / Seaborn	For visualizing training metrics, infection progression, and model evaluation results (backend/dashboard side).

3.4 SOFTWARE SOLUTION

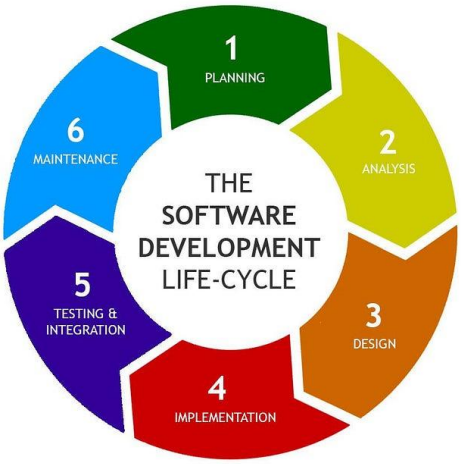

Figure 6: SDLC Life Cycle

We will follow the Agile Software Development Life Cycle (SDLC) model to guide our project. Agile provides a flexible and iterative approach, allowing us to plan, develop, test, and deliver high-quality software in stages. This methodology encourages continuous improvement, frequent user feedback, and quick adjustments, ensuring the final product meets user needs effectively. By breaking the project into manageable sprints, the team can focus on delivering specific features incrementally, leading to a more efficient and responsive development process. This approach will help us build a robust and user-friendly land management application.

4. PROJECT REQURMENTS

4.1 FUNCTIONAL REQUIREMENTS

- Allow users to capture and upload images of cinnamon leaves through the mobile app.
- Preprocess images (resize, denoise, and segment infected regions) before analysis.
- Annotate infection stages with severity percentage and maintain infection progression history.
- Extract temporal and spatial features from images to support time-series analysis.
- Apply trained ML models (LSTM/GRU) to forecast infection spread over a defined future period (e.g., 3 days).
- Display infection severity (current and predicted) with visualization graphs/charts.
- Provide farmers with early treatment recommendations (e.g., fungicide spraying, pruning, isolating plants).
- Store farmer usage history and infection prediction records securely.

4.2 NON-FUNCTIONAL REQUIREMENTS

- **Accuracy** – Achieve >85% accuracy for infection prediction using MAE, RMSE, and R^2 as evaluation metrics.
- **Usability** – Provide a farmer-friendly interface with simple navigation and local language support.
- **Performance** – Predictions must be generated within 5 seconds per image input.
- **Reliability** – Maintain 99% uptime with effective error handling and recovery.
- **Security** – Ensure user data and images are protected via encryption and authentication.
- **Scalability** – The backend must support multiple farmers and high image upload volumes simultaneously.
- **Portability** – The application should run on Android and iOS devices without loss of functionality.
-

4.3 USER REQUIREMENTS

- Farmers should be able to easily capture leaf images and upload them without complex steps.
- The app should present both current infection status and predicted spread clearly using visualizations.
- Farmers must receive actionable advice in simple, non-technical language.
- Historical records should be viewable to compare predictions with actual outcomes.
- The app should provide multi-language support (Sinhala, Tamil, English) for wider accessibility.

4.4 HARDWARE REQUIREMENTS

- Smartphone with minimum 2GB RAM and 32GB storage.
- Camera: ≥8MP for clear leaf image capture.
- Internet: 4G/5G or Wi-Fi connectivity.

4.5 SOFTWARE REQUIREMENTS

- Mobile Application (Android Studio / Flutter / React Native).
- Backend API (Python Flask / FastAPI).
- Image processing: OpenCV, TensorFlow/Keras, or PyTorch.
- Database: Firebase or SQL-based storage.

4.6 TEST CASES

Table IV: Test Cases

Test Scenario	**Image Upload & Preprocessing**
Test Scenario ID	**T1**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that the system allows a farmer to capture and upload a cinnamon leaf image, and preprocesses it (resize, denoise, segment).	The user must have a mobile device with the application installed. The user is logged into the app. The device camera is functional and has permission enabled.	User captures and uploads a cinnamon leaf image.	The system resizes, denoises, and segments the infected regions. Preprocessed image is stored and ready for feature extraction.	Image is added to dataset. user receives confirmation of successful upload.

Test Scenario	**Infection Prediction using ML Model**
Test Scenario ID	**T2**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that the system applies the LSTM/GRU model to predict future spread of fungal infection.	A preprocessed leaf image must be available. ML model must be trained and deployed.	User submits a preprocessed image for prediction.	The model outputs the current infection percentage. The model forecasts infection spread (e.g., 3 days later).	Prediction is stored in the database with timestamp. Prediction is displayed on user's dashboard.

Test Scenario	**Treatment Recommendation**
Test Scenario ID	**T3**

Description	Pre - Conditions	Inputs	Expected Results	Post-conditions
Verify that the system provides early treatment recommendations based on predicted infection severity.	Prediction results must be available. Recommendation rules engine must be configured.	Predicted infection percentage (e.g., 45% → high risk).	The app displays tailored recommendations (e.g., spray fungicide, prune infected leaves).	User can save or dismiss recommendations. Recommendations are stored in history for reference.

5. DESCRIPTION OF PERSONAL AND FACILITIES

Table V: Description of personal and facilities

Registration No	Name	Task Description
IT22578778	Randeni R.P.R.M	• Collect and preprocess cinnamon leaf images. • Annotate infection stages and build dataset. • Train LSTM/GRU models for fungal spread prediction. • Develop mobile app for image input and prediction output. • Implement visualization and treatment recommendation features. • Validate system accuracy with field testing.

6. GANT CHART

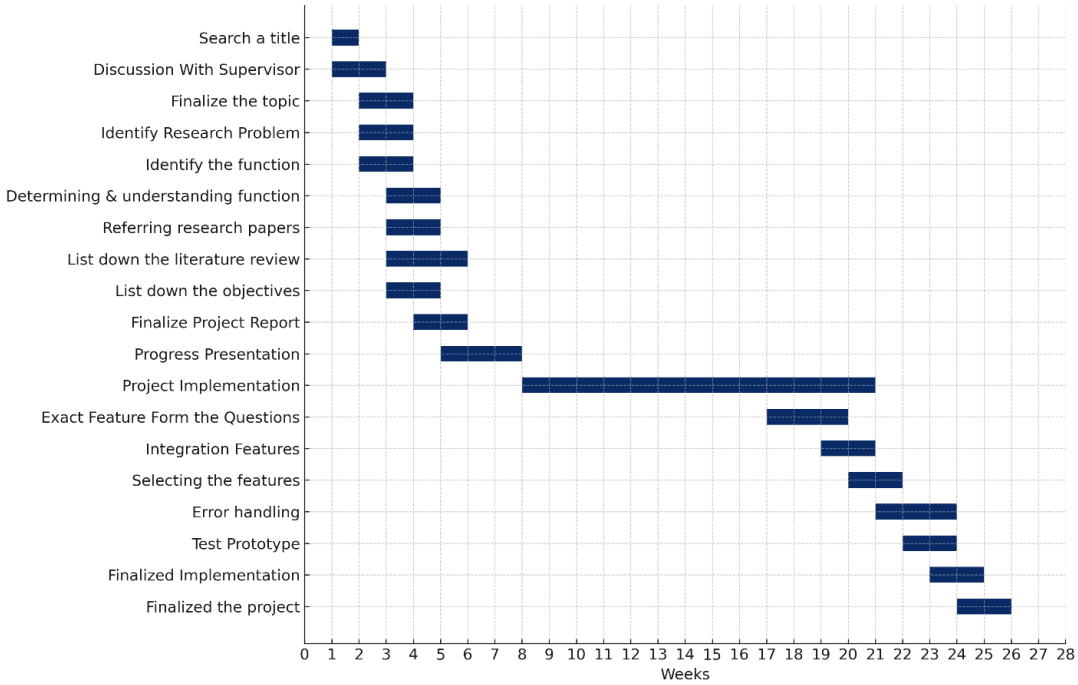

Figure 7:Gant Chart

7. WORK BREAKDOWN CHART

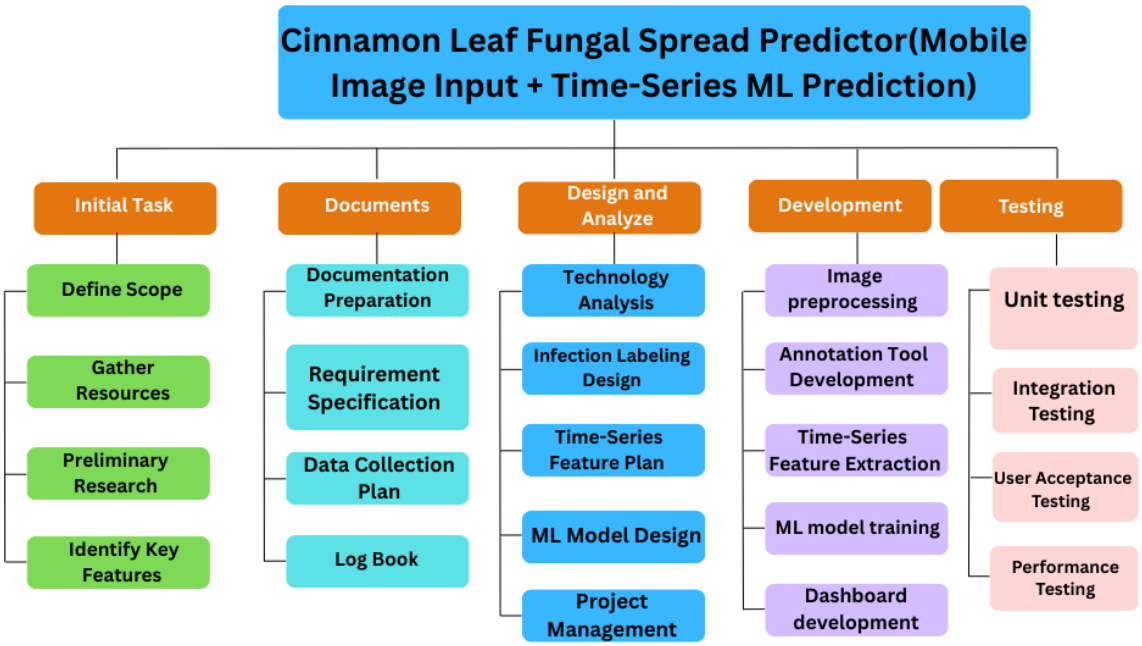

Figure 8: WBS Diagram

8. COMMERCIALIZATION PLAN

8.1 BUSINESS PLAN

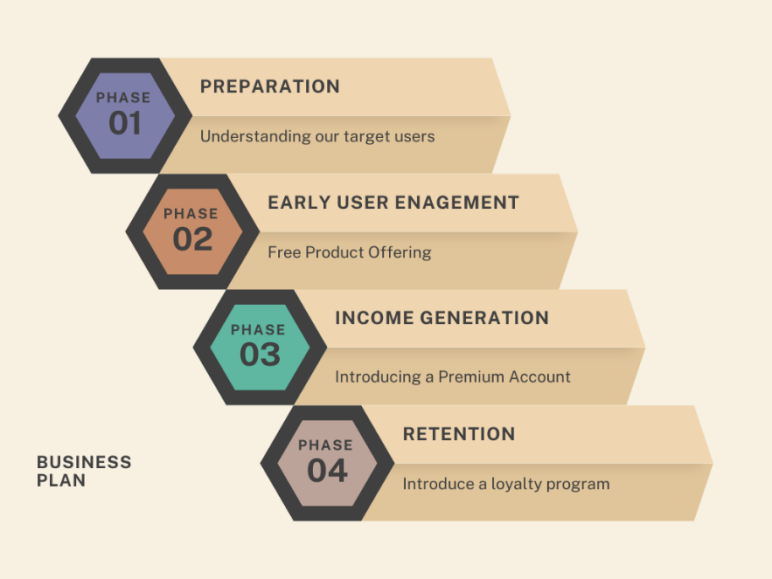

Figure 9: Business Plan

Phase 01 – Preparation
Identify target users (cinnamon farmers), collect datasets, and ensure the application and ML models function reliably.

Phase 02 – Early User Engagement
Offer the mobile app free to early users for feedback on usability and prediction accuracy, helping refine the system.

Phase 03 – Income Generation
Introduce a subscription model with basic free features and affordable premium plans (e.g., 400–500 LKR/month) for advanced forecasting.

Phase 04 – Retention
Implement loyalty and referral programs with discounts and bonus features to retain long-term users and expand adoption.

8.2 MARKETING PLAN

SOCIAL MEDIA MARKETING

We will use social media platforms to create awareness about the Cinnamon Leaf Fungal Spread Predictor. Our focus will be on educating farmers about the importance of early fungal detection and prediction. We will create simple posts, short videos, and success stories from farmers who benefit from the app. The campaign will mainly target Facebook, WhatsApp, and YouTube, as these platforms are widely used by farming communities.

SALES APPROACH

Our team will directly connect with cinnamon farmers and plantation managers through agricultural cooperatives, farmer associations, and training workshops. We will also participate in local agricultural exhibitions and online webinars to demonstrate the app's features and how it can help prevent crop losses.

CREATING CUSTOMER PACKAGES

We will introduce flexible subscription plans to suit different farmers. A free version will provide basic infection detection, while premium plans will include advanced forecasting, historical analysis, and treatment recommendations. Affordable pricing will encourage small- and medium-scale farmers to upgrade, while loyalty rewards will motivate them to stay long-term.

9. BUDGET AND BUDGET JUSTIFICATION

Table VI:Budget and Budget Justification

Description	Amount (LKR)
IoT Sensors (NPK JXBS-3001, Moisture FC-28, Temp/Humidity DHT11, pH/EC Probes)	40,000
ESP32 Microcontrollers + Wi-Fi/LoRa Modules	15,000
Cloud Hosting & APIs (Azure, MQTT broker, Firebase/MongoDB Atlas)	25,000
ML/AI Model Training Infrastructure (Cloud GPU credits / Colab Pro / Azure ML)	20,000
Drone/Satellite Imagery (GIS Yield Planner)	10,000
Mobile Application Development (Fungal Spread Predictor)	Free
Multi-Sensor Array for Post-Harvest Quality Guard (Temp, RH, CO_2, O_2, VOC, Light, Airflow, Dust, Vibration, Surface Moisture)	50,000
Dev Tools & Software (VS Code, Postman, GitHub, GIS software licenses, testing tools)	Free
Miscellaneous (Travel to plantations, field trials, calibration, replacements)	10,000
Total Estimated Budget	170,000

10. CONCLUSION

The Cinnamon Leaf Fungal Spread Predictor provided here is a novel, mobile-phone technology bringing a innovative solution to the daunting challenge of fungal infection in cinnamon cultivation. Using image processing, machine learning, and mobile technology, the system enables farmers to skip from traditional visual inspection to a data-driven, preventive approach to disease management.

The domain-specific aspect offers end-to-end support through the following contributions:

- Image Preprocessing and Annotation ensures accurate dataset preparation by resizing, denoising, and segmenting infected parts of cinnamon leaves.
- Temporal Data Modeling transforms annotated infection stages into structured datasets that are usable for forecasting fungal growth.
- LSTM and GRU Prediction Models provide authentic predictions of infection spread within a given time period, giving farmers an early warning.
- Mobile Application Integration offers infection status, prediction, and treatment recommendations to farmers in a usable and intuitive frontend.

Collectively, all the innovations facilitate farmers to get early warnings and actionable information, thus allowing them to take preventative action through pruning, spraying, or quarantining infected plants before they can cause extensive crop loss. The system directly supports yield protection, cost savings, and environmentally sustainable cinnamon farming practices in Sri Lanka.

This paper demonstrates how smart agriculture solutions can transform disease control from reactive treatment to proactive prevention. Next steps can include integration with climatic data for improved forecasting, expansion of the fungal disease coverage to more Cylindrocladium and Anthracnose crops, and use of federated learning for model training on a large scale without jeopardizing farmer data privacy.

The Cinnamon Leaf Fungal Spread Predictor is not only a novel entry in precision agriculture but also a cost-effective, scalable, and farmer-friendly technology that reinforces Sri Lanka as a world power in high-quality cinnamon production.

11. REFERENCE LIST

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12. APPENDICIES

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Figure 10:Appendices I